

# Pranav Mallela

Software Engineer • ML Engineer

pmallela@umich.edu • 734 913 3384 • US Citizen • linkedin.com/in/pranav-mallela • Ann Arbor, MI

## EDUCATION

### University of Michigan

Ann Arbor, MI

Computer Science and Engineering **M.S.**

Aug. 2024 – May 2026

Coursework: Self-Driving Cars; Course Assistant: Data Manipulation and Analysis, Database Management Systems

### Indian Institute of Technology (IIT)

Ropar, India

Computer Science and Engineering **B.Tech.** 8.77 GPA

Nov. 2020 – May 2024

Coursework: Data Structures, Algorithms, Databases, Software Engineering, Machine Learning, Data Science

TA: Guided multiple student groups in development projects based on stakeholder requirements in the course CP301.

## SKILLS & CERTIFICATIONS

*Programming Languages:* C++, Python (PyTorch), C, JavaScript, Java, SQL and MATLAB.

*Development:* HTML, CSS, React.js, Node.js, RESTful APIs, MongoDB, PostgreSQL, Flutter, Firebase, GCP, Git.

*Publications:* Paper accepted at ECAI 2024 in Human-AI Teams, Currently targeting AAMAS 2025 in Fairness.

*Google Certification:* Cloud Digital Leader Pathway and Cloud Engineer Pathway.

## WORK EXPERIENCE

### MathWorks

Hyderabad, India

Engineering Development Group Intern

May 2023 – July 2023

- Revamped 150+ tests to efficiently exercise overlapping software functionality.
- Reduced technical debt by a substantial 2600 lines. Realized up to 80% reduction in test execution time.
- Leveraged object-oriented software design to develop a testing framework which enabled backward compatibility.
- Spearheaded a team of 5 to build an interview platform in the Summer Hackathon and secured the 2<sup>nd</sup> position.

## RESEARCH EXPERIENCE

### Michigan AI Lab

Ann Arbor, MI

Student Researcher, University of Michigan

July 2024 - Present

- Formulated a novel approach to tackle selection bias in public health datasets.
- Incorporated selection mechanisms into an agent-based model for the simulation of infection spread disparity.

### Game Theory and Machine Learning Lab

Ropar, India

Student Researcher, IIT Ropar

Dec. 2022 – July 2024

- Proposed a probabilistic estimation of human labels, reducing the cost of human input by a factor of 8.
- Authored a paper on a Human-AI framework which was accepted at the leading AI conference in Europe.
- Analyzed multiple image and natural language datasets to train a neural network for multi-way classification.

## PROJECT EXPERIENCE

### Academic Management System

April 2023

Java, JUnit, PostgreSQL

- Modeled academic processes through UML diagrams; Developed the system guided by Java Object Oriented Principles.
- Developed a suite of 100 unit tests using JUnit Jupiter; Achieved 96% line coverage and 89% branch coverage.

### Recruiter Portal for IIT Ropar

August 2023

React, Django, PostgreSQL

- Digitalized job and internship details submission, enhancing recruiter engagement from multiple companies.

### Hospital Management System

January 2023

React, Firestore - NoSQL, Google Cloud

- Built and containerized a serverless application using Docker and Cloud Build, then deployed it to Google Cloud Run.